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Cloud Storage Technologies: Object, Block, and File Storage Compared

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Abstract

- Cloud storage eliminates the need for expensive physical storage devices.
- Different storage technologies are designed for different workloads.
- Understanding storage types helps improve application performance.
- Organizations use cloud storage to increase scalability and reduce costs.
- This presentation compares architecture, performance, scalability, cost, security, and real-world applications.

Introduction

- Data is stored in remote cloud data centers.
- Accessible from anywhere using an internet connection.
- Cloud providers maintain hardware and infrastructure.
- Storage resources can be increased or decreased on demand.
- Offers high availability, redundancy, backup, and disaster recovery.

Object Storage

- Stores data as independent objects with unique identifiers.
- Includes rich metadata for efficient search and management.
- Provides virtually unlimited scalability.
- Highly durable through data replication.
- Best for unstructured data, backups, archives, media, and cloud-native applications.

Block Storage

- Divides data into fixed-size blocks.
- Provides high performance and low latency.
- Ideal for databases and virtual machines.
- Supports high IOPS for enterprise workloads.
- Frequently used in transactional systems.

File Storage

- Organizes data in folders and files.
- Easy to manage and share.
- Suitable for shared documents and network drives.
- Examples: NFS, SMB, Amazon EFS.

Comparison

- Feature | Object | Block | File

Scalability: High | Medium | Medium

Performance: Medium | High | Medium

Structure: Objects | Blocks | Files/Folders

Best Use: Media | Databases | Shared Files

Advantages & Limitations

- Object: +Scalable, -Higher latency
- Block: +Fast, -Complex management
- File: +Easy sharing, -Limited scalability

Applications

- Object: Backup, AI datasets, videos.
- Block: Databases, ERP, VMs.
- File: Collaboration, home directories, content management.



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Conclusion

- Cloud storage is a fundamental technology for modern businesses and cloud computing.
- **Object Storage** is ideal for storing large volumes of unstructured data such as images, videos, backups, and IoT data due to its excellent scalability and cost-effectiveness.
- **Block Storage** provides high performance and low latency, making it the best choice for databases, and mission-critical enterprise applications.

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THANK YOU